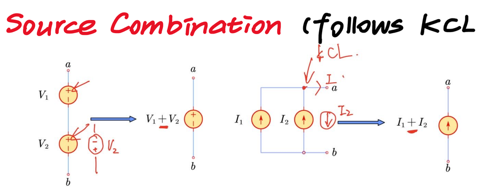
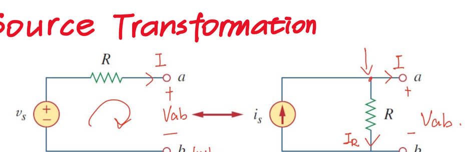
ECE 211

2026年3月13日 星期五 09:05

Source Combination (follows KCL & KVL)



Source Transformation



KVL: $V_s = I \cdot R + V_{ab}$

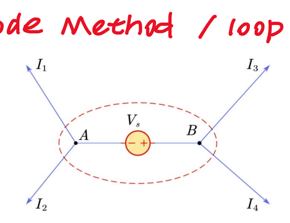
KCL: $i_s = I + \frac{V_{ab}}{R}$ ⇓ $i_s \cdot R = I \cdot R + V_{ab}$

Same Current-Voltage Relation at a-b terminals

$v_s = i_s R$ or $i_s = \frac{v_s}{R}$

Equivalent Circuit
a circuit having similar i-v characteristics

Node Method / loop Current Method



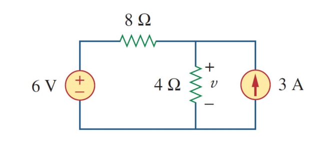
Find Reference → KCL & KVL
(assume / ⏚)

$\begin{cases} V_B - V_A = V_s \\ I_1 + I_2 + I_3 + I_4 = 0 \end{cases}$

Superposition (use linearity)

$\begin{cases} \text{Current Source} \to \text{Open loop} \\ \text{Voltage Source} \to \text{Short Circuit} \end{cases}$

Use superposition to determine v



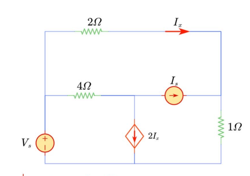
$v = V_1 + V_2$ from ① 6V → from ① 3A

1. To obtain V_1, turn off current source (open circuit)
KVL: $12 i_1 - 6 = 0$
$i_1 = 0.5 A$
⇒ $V_1 = 4 i_1 = 2V$

2. To obtain V_2, turn off voltage source (short circuit)
CDR: $i_3 = \frac{8}{4+8} \times 3 = 2A$
$V_2 = 4 i_3 = 8V$

$V = V_1 + V_2 = 10V$.

Use superposition to determine I_x in this circuit



$I_x = k_1 V_s + k_2 I_s$ (I_{x1}, I_{x2})

1. keep V_s, $I_s = 0$ open cbt. I_{x1}
KVL: $V_s - 2I_{x1} - 1 \cdot I_{x1} = 0$ ⇒ $I_{x1} = \frac{1}{3} V_s$

2. keep I_s, $V_s = 0$
CDR: $I_{x2} = -\frac{1}{1+2} \cdot I_s = -\frac{1}{3} I_s$

∴ $I_x = I_{x1} + I_{x2} = \frac{1}{3} V_s - \frac{1}{3} I_s$

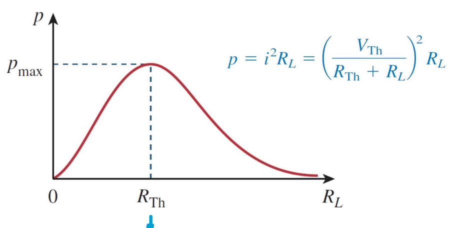
Thevenin & Norton Equivalent

Find $\begin{cases} V_T \to \text{Open Circuit} \\ I_N \to \text{Short Circuit} \end{cases}$

$R_T = \frac{V_T}{I_N}$

Maximum Power Transfer

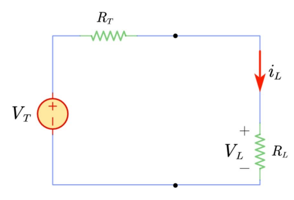
Power delivered to the load as a function of R_L



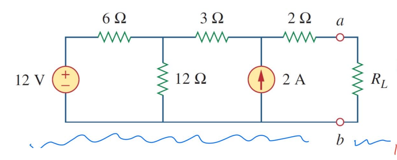
$p = i^2 R_L = \left(\frac{V_{Th}}{R_{Th} + R_L}\right)^2 R_L$

$\frac{dP}{dR_L} = 0 \Rightarrow R_L = R_T$ $P_a = \frac{V_T^2}{4R_T}$

What is the maximum power a load can get from the following circuit?



Find R_L for maximum power transfer.



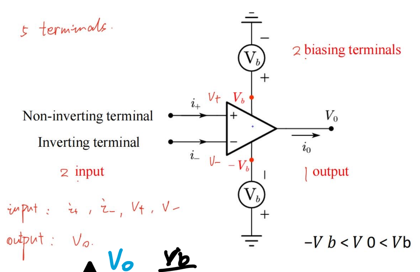
① R_T. ⇒ Remove all indep. sources, V_s → short, I_s → open. $R_L = R_T$
$R_T = R_{eq} = 2 + 3 + 6 // 12 = 9\Omega$

② V_T ⇒ open cbt a~b. open. $V_{ab} = V_T = V_{2A}$.
$i_2 = -2A$.
KVL for loop 1: $-12 + 6i_1 + 12(i_1 - i_2) = 0$ ⇒ $i_1 = \frac{-2}{3} A$
$-12 + 6i_1 + 3i_2 + V_T = 0$ ⇒ $V_T = 22V$. ∴ $P_a = \frac{V_T^2}{4R_T} = 13.44 W$

Signal Processing Circuit

Operational Amplifier (Op-Amp)

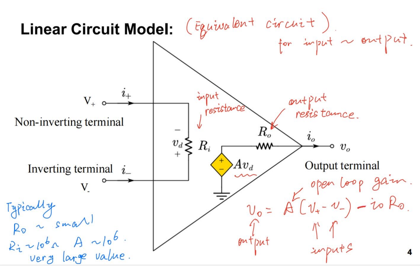
5 terminals.



input: i_+, i_-, V_+, V_-
output: V_o

$-Vb < V0 < Vb$

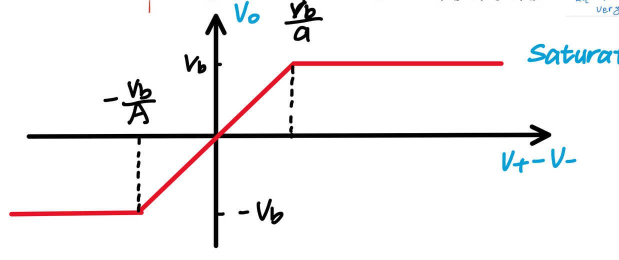
Linear Circuit Model: (Equivalent circuit for input ~ output.)



Typically R_o ~ small, $R_i \sim 10^6 \Omega$, $A \sim 10^6$ very large value.

$V_o = A(V_+ - V_-) - i_o R_o$

(if R_o small → 0) $V_o = A(V_+ - V_-)$



Saturation Region (non-linear)

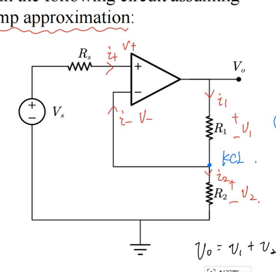
Linear:
$|V_+ - V_-| < \frac{V_b}{A}$

if $A \to \infty$ → $V_+ = V_-$, $i_+ = i_- = 0$

Ideal Op-Amp:
- Infinite open-loop gain ($A \to \infty$)
- Infinite input Resistance ($R_i \to \infty$)
- Zero Output Resistance ($R_o \to 0$)

Examples:

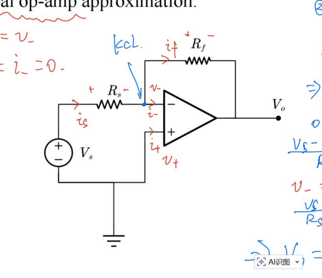
Obtain V_o in the following circuit assuming ideal op-amp approximation:



① $V_+ = V_-$, $i_+ = i_- = 0$.
② KCL: $i_1 = i_- + i_2$, $i_1 = i_2$.
③ $V_2 = V_- = V_+ (i_+ = 0) = V_s$
⇒ $i_2 = \frac{V_2}{R_2} = \frac{V_s}{R_2}$ = i_1
$V_o = V_1 + V_2 = R_1 i_1 + R_2 i_2 = \frac{R_1 + R_2}{R_2} V_s$ > 0.

Note: Output is connected to the inverting input

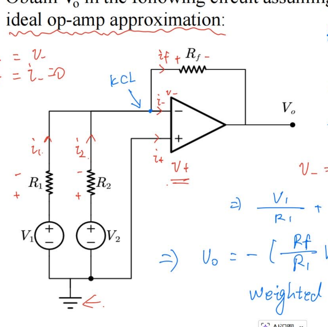
Obtain V_o in the following circuit assuming ideal op-amp approximation:



① $V_+ = V_-$, $i_+ = i_- = 0$.
② KCL: $i_s = i_f + i_- = 0$ ⇒ $i_s = i_f$.
ohm's law. $\frac{V_s - V_-}{R_s} = \frac{V_- - V_o}{R_f}$
$V_- = V_+ = 0$. $\frac{V_s}{R_s} = \frac{-V_o}{R_f}$
⇒ $V_o = -\left(\frac{R_f}{R_s}\right) V_s$ ≤ 0.

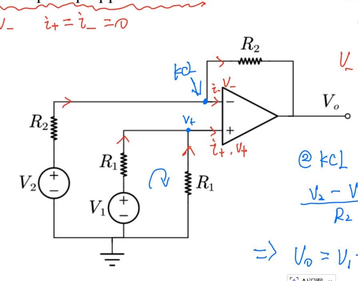
Note: Output is connected to the inverting input

Obtain V_o in the following circuit assuming ideal op-amp approximation:



① $V_+ = V_-$, $i_+ = i_- = 0$
KCL: $i_1 + i_2 = i_- + i_f$
$i_1 + i_2 = i_f$.
ohm's law. $\frac{V_1 - V_-}{R_1} + \frac{V_2 - V_-}{R_2} = \frac{V_- - V_o}{R_f}$
$V_- = V_+ = 0$.
⇒ $\frac{V_1}{R_1} + \frac{V_2}{R_2} = \frac{-V_o}{R_f}$
⇒ $V_o = -\left(\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2\right)$
weighted sum of input voltages

Obtain V_o in the following circuit assuming ideal op-amp approximation:



$V_+ = V_-$, $i_+ = i_- = 0$
① Loop on "+" side. VDR: $V_+ = V_+ = V_{R_1} = \frac{R_1}{R_1 + R_1} V_1 = \frac{V_1}{2}$
② KCL at "−" side. $\frac{V_2 - V_-}{R_2} = \frac{V_- - V_o}{R_2}$
⇒ $V_o = V_1 - V_2$.

Ideal Circuit elements

C: $i = C\frac{dv}{dt}$.
⇒ $V(t) = \frac{1}{C}\int_{-\infty}^{t} i(x)dx$
$= \frac{1}{C}\int_{-\infty}^{0} i(x)dx + \frac{1}{C}\int_{0}^{t} i(x)dx$
$= V_0$ initial state.

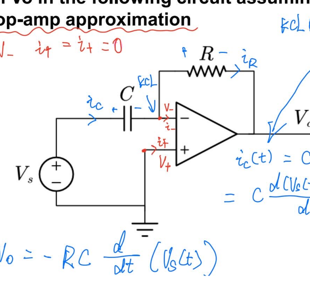
$i = C\frac{dv}{dt}$
$V(t) = \frac{1}{C}\int_{-\infty}^{t} i(x)dx$

L: $V = L\frac{di}{dt}$
⇒ $i(t) = \frac{1}{L}\int_{-\infty}^{t} V(x)dx$
$= \frac{1}{L}\int_{-\infty}^{0} V(x)dx + \frac{1}{L}\int_{0}^{t} V(x)dx$
$= i_0$ "initial state"

$V = L\frac{di}{dt}$
$I(t) = \frac{1}{L}\int_{-\infty}^{t} V(x)dx$

Voltage Differentiator

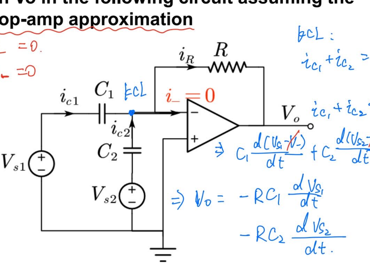
Obtain Vo in the following circuit assuming the ideal op-amp approximation



$V_+ = V_- = 0$, $i_+ = i_+ = 0$
KCL @ "−": $i_C = i_- + i_R = i_R$
$i_C(t) = C\frac{dV_C}{dt} = C\frac{d(V_s(t) - V_-)}{dt} = \frac{V_- - V_o}{R}$
⇒ $V_o = -RC\frac{d}{dt}(V_s(t))$

Adder Differentiator

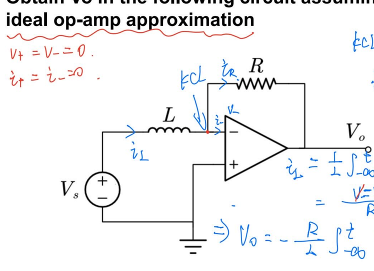
Obtain Vo in the following circuit assuming the ideal op-amp approximation



$V_+ = V_- = 0$. $i_+ = i_- = 0$
KCL: $i_{C_1} + i_{C_2} = i_- + i_R$.
$i_{C_1} + i_{C_2} = i_R$.
$C_1\frac{d(V_{s1} - V_-)}{dt} + C_2\frac{d(V_{s2} - V_-)}{dt} = \frac{V_- - V_o}{R}$
⇒ $V_o = -RC_1\frac{dV_{s1}}{dt} - RC_2\frac{dV_{s2}}{dt}$.

Voltage Integrator

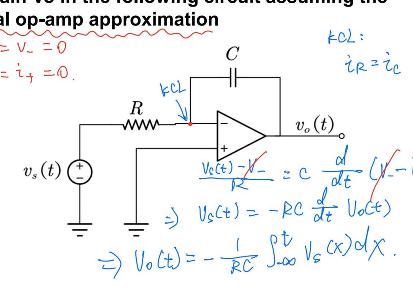
Obtain Vo in the following circuit assuming the ideal op-amp approximation



$V_+ = V_- = 0$. $i_+ = i_- = 0$.
KCL @ "−": $i_L = i_- + i_R$. ⇒ $i_L = i_R$.
$i_L = \frac{1}{L}\int_{-\infty}^{t} V_L(x)dx$ $(V_s - V_-)$
$= \frac{V_- - V_o}{R} = i_R$.
⇒ $V_o = -\frac{R}{L}\int_{-\infty}^{t} V_s(x)dx$

Integrator

Obtain Vo in the following circuit assuming the ideal op-amp approximation



$V_+ = V_- = 0$. $i_- = i_+ = 0$.
KCL: $i_R = i_C$
$\frac{V_s(t) - V_-}{R} = C\frac{d}{dt}(V_- - V_o(t))$
⇒ $V_s(t) = -RC\frac{d}{dt}V_o(t)$
⇒ $V_o(t) = -\frac{1}{RC}\int_{-\infty}^{t} V_s(x)dx$.

Linearity

Recall from previous example: $V_O(t) = V_{O(t=0)} - \frac{1}{RC}\int_0^t v_S(t)dt$
(Initial State)

Input: $v_S(t)$

Output (Total Response): $V_O(t) = V_{O(t=0)} - \frac{1}{RC}\int_0^t v_S(t)dt$

Zero State Response: $V_{O,ZSR}(t) = -\frac{1}{RC}\int_0^t v_S(t)dt$ Y_{ZS}.

Zero Input Response: $V_{O,ZIR}(t) = V_{O(t=0)}$ Y_{ZI}

zero-state linear system $y_{ZS}(t)$.
$f_1(t) \to \square \to y_{ZS,1}(t)$
$f_2(t) \to \square \to y_{ZS,2}(t)$
⇒ $f_3(t) \to \square \to y_{ZS,3}(t)$
$= k_1 f_1(t) + k_2 f_2(t)$ $= k_1 y_{ZS,1}(t) + k_2 y_{ZS,2}(t)$.

zero input linear system $y_{ZI}(t)$.
$f_1(0) \to \square \to y_{ZI,1}(t)$
$f_2(0) \to \square \to y_{ZI,2}(t)$
⇒ $f_3(0) \to \square \to y_{ZI,3}(t)$
$= k_1 f_1(0) + k_2 f_2(0)$ $= k_1 y_{ZI,1}(t) + k_2 y_{ZI,2}(t)$

Determine if the system is linear or not

1). zero input. $y_{ZI}(t) = y_0 e^{-t}$. zero-input Linear.
$y_{0,1} \to \square \to y_{ZI,1}(t) = y_{0,1}e^{-t}$
$y_{0,2} \to \square \to y_{ZI,2}(t) = y_{0,2}e^{-t}$
$y_{0,3} \to \square \to y_{ZI,3}(t) = y_{0,3}e^{-t}$
$= k_1 y_{0,1} + k_2 y_{0,2}$ → $= (k_1 y_{0,1} + k_2 y_{0,2})e^{-t}$
$= k_1 y_{0,1}e^{-t} + k_2 y_{0,2}e^{-t}$
$= k_1 y_{ZI,1}(t) + k_2 y_{ZI,2}(t)$

2). zero state
$y_{ZS}(t) = -\int_0^t f(x)dx$
$f_1(t) \to \square \to y_{ZS,1}(t) = -\int_0^t f_1(x)dx$
$f_2(t) \to \square \to y_{ZS,2}(t) = -\int_0^t f_2(x)dx$
$f_3(t) \to \square \to y_{ZS,3}(t) = -\int_0^t f_3(x)dx$.
$= k_1 f_1(t) + k_2 f_2(t)$ → $= -\int_0^t (k_1 f_1(x) + k_2 f_2(x))dx$
$= -\int_0^t k_1 f_1(x)dx - \int_0^t k_2 f_2(x)dx$

zero-state linear
$= -k_1\int_0^t f_1(x)dx - k_2\int_0^t f_2(x)dx$
$= k_1 y_{ZS,1}(t) + k_2 y_{ZS,2}(t)$.

Linear System!

Time Invariance

Delayed inputs cause equally delayed zero-state outputs

t_d.

if

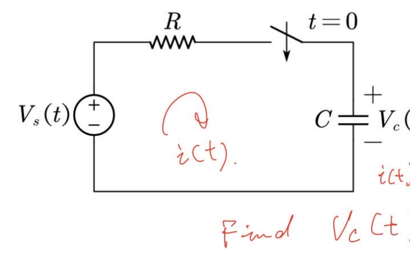
$\mathbf{f_1(t)} \to \boxed{\text{time-invariant}} \to \mathbf{y_1(t)}$,

then

$\mathbf{f_2(t) = f_1(t - t_d)} \to \boxed{\text{time-invariant}} \to \mathbf{y_2(t) = y_1(t - t_d)}$, TI.

$y_2(t) \neq y_1(t - t_d)$ → × TI.

First-Order RC Circuit



$i(t) = C\frac{dV_C(t)}{dt}$.

Find $V_C(t)$.

$V_s(t) - RC\frac{dV_C(t)}{dt} - V_C(t) = 0$

$\frac{dV_C(t)}{dt} + \frac{1}{RC}V_C(t) = \frac{1}{RC}V_s(t)$ → ODE

$f(t) = \frac{dy}{dt} + ay(t)$

- How to solve?
  - Start with a simple case: constant input

$\boxed{f(t) = K}$ ⇒ $K = \frac{dy}{dt} + ay(t)$ ⇒ Find $y(t)$.

1). Assume $y(t) = B$. (constant) → find B = ?
plug into ODE: $K = \frac{dB}{dt} + aB$, $\frac{dB}{dt} = 0$.
⇒ $K = aB$ ⇒ $B = \frac{K}{a}$
$y(t) = \underbrace{\frac{K}{a}}_{y_p(t)} + \underbrace{\cdots}_{y_h(t)}$

2). $y(t) = y_p(t) + y_h(t)$ homogenous solution, particular solution
plug into ODE
$K = \frac{d}{dt}\left(\frac{K}{a} + y_h(t)\right) + a\left(\frac{K}{a} + y_h(t)\right)$.
⇒ $\cancel{K} = \frac{d}{dt}y_h(t) + \cancel{K} + a y_h(t)$
⇒ $\frac{d}{dt}y_h(t) + a y_h(t) = 0$.
guess $y_h(t) = Ae^{-at}$ ⇒ Find A
$[-aAe^{-at} + a \cdot Ae^{-at} = 0]$

Initial condition. $V_C(0^-) = V_C(0^+) = V_C(0)$
$y(0^-) = y(0^+) = y(0)$
at $t=0$. $y(t=0) = \frac{K}{a} + Ae^{-a \cdot 0}$
$= \frac{K}{a} + A = y(0^-)$
⇒ $A = y(0^-) - \frac{K}{a}$
∴ $y(t) = \frac{K}{a} + \left(y(0^-) - \frac{K}{a}\right)e^{-at}$, $t > 0$

$$\mathbf{K} = \frac{\mathbf{dy}}{\mathbf{dt}} + \mathbf{ay(t)}$$

- For $t > 0$,

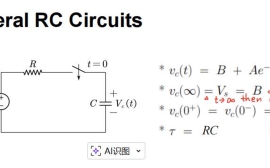
$$\mathbf{y(t) = y_p(t) + y_h(t) = B + Ae^{-at} = \frac{K}{a} + \left(y(0^+) - \frac{K}{a}\right)e^{-at}}$$

$= \underbrace{(1 - e^{-at})\frac{K}{a}}_{y_{ZS}} + \underbrace{y(0^+)e^{-at}}_{y_{ZI}}$

$\frac{V_s}{RC} = \frac{dV_C(t)}{dt} + \frac{1}{RC}V_C(t)$
$K = \frac{dy}{dt} + a\,y(t)$
$y(t) = \underbrace{\frac{K}{a}}_{y_p} + \underbrace{\left(y(0) - \frac{K}{a}\right)e^{-at}}_{y_h}$.
⇒ $y_p = \frac{K}{a} = \frac{V_s/RC}{1/RC} = V_s$.
$y_h = (y(0^+) - V_s)e^{-at}$
∴ $V_C(t) = V_s + (V_C(0^+) - V_s)e^{-at}$
for $t > 0$

A Simple Solution Method

General RC Circuits



- $v_c(t) = B + Ae^{-\frac{t}{\tau}}$ $V_C(\infty) = B + A\cancel{e^{-\infty}}$
- $v_c(\infty) = V_s = B$ ← $t \to \infty$ then C behave like an open cbt
- $v_c(0^+) = v_c(0^-) = v_0 = B + A$
- $\tau = RC$

$A = V_0 - B$.

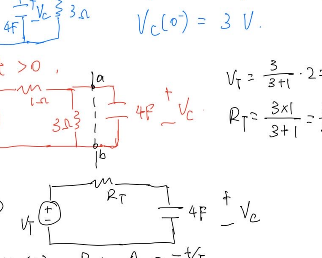
$v_c(t) = V_s + (v_c(0^-) - V_s)e^{-\frac{t}{RC}}$ for $t > 0$

Example ↓

Consider the circuit below.

Assume the switch has been in position A for a long time and it switches to position B at time **t = 0**.

Determine $V_{ZS}(t)$, $V_{ZI}(t)$, $i_{ZS}(t)$, $i_{ZI}(t)$ for capacitor.



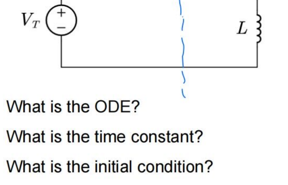
① $t < 0$, long time
cap. 4F ⇒ open cbt
$V_C(0^-) = 3V$.

② $t > 0$,
$V_T = \frac{3}{3+1} \cdot 2 = \frac{3}{2}V$
$R_T = \frac{3 \times 1}{3+1} = \frac{3}{4}\Omega$

③ $V_C(t) = B + Ae^{-t/\tau}$
$\tau = R_T C = \frac{3}{4} \cdot 4 = 3$ [s].
$V_C(\infty) = B = V_T = \frac{3}{2}$ [V] (open cbt for C)
$V_C(0) = V_C(0^-) = 3V$
$= A + B$ ⇒ $A = \frac{3}{2}$.
⇒ $V_C(t) = \frac{3}{2} + \frac{3}{2}e^{-t/3}$ [V].
$= \frac{3}{2} + \left(V_C(0^-) - \frac{3}{2}\right)e^{-\frac{t}{3}}$.
∴ $V_{ZS} = \frac{3}{2} - \frac{3}{2}e^{-t/3}$ [V] ($V_C(0^-) = 0$)
$V_{ZI} = V_C(0^-)e^{-\frac{t}{3}} = 3e^{-t/3}$ [V] ($V_T = 0$)

$V_C(t) = V_T + (V_C(0^-) - V_T)e^{-t/\tau}$
$V_{ZS}(t) = V_T - V_T e^{-t/\tau}$
$V_{ZI}(t) = V_C(0^-)e^{-t/\tau}$.

First Order RL → Thevenin to Norton



What is the ODE?

What is the time constant?

What is the initial condition?

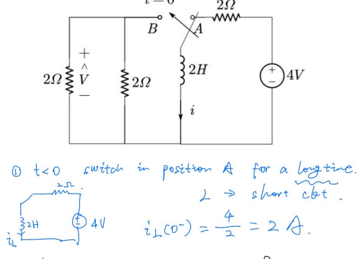
Inductor: $V_L(t) = L\frac{di_L(t)}{dt}$.
KCL @ node ①
$I_N = \frac{V_L}{R_T} + i_L$
$= \frac{L}{R_T}\frac{di_L(t)}{dt} + i_L$
⇒ $\frac{di_L}{dt} + \frac{R_T}{L}i_L = \frac{R_T}{L}I_N$, $t > 0$.
first order ODE with constant source and constant coeff.
$\frac{dy}{dt} + ay = K$.

Solution $i_L(t) = \underbrace{B}_{y_p} + \underbrace{Ae^{-t/\tau}}_{y_h}$, $t > 0$
1. $\tau = \frac{L}{R_T}$ (time constant)
2. $i_L(\infty) = B = I_N$. (L → short cbt.)
3. $i_L(0^+) = B + A = i_L(0^-)$ given
⇒ $A = i_L(0^-) - I_N$.
∴ $i_L(t) = I_N + (i_L(0^-) - I_N)e^{-t/\tau}$.

Consider the circuit below.

Assume the switch has been in position A for a long time and it switches to position B at time **t = 0**.

Determine $\mathbf{i_{ZI}(t)}$ and $\mathbf{i_{ZS}(t)}$.



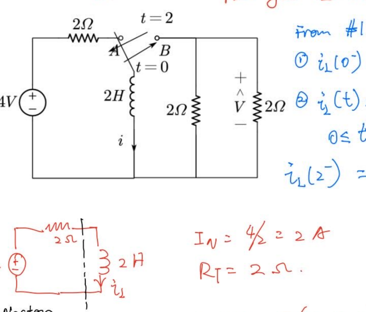
① $t < 0$ switch in position A for a long time. L → short cbt.
$i_L(0^-) = \frac{4}{2} = 2A$.

② $t > 0$. Find I_N, R_T
$I_N = 0$. (no indep. source)
$R_T = 2\Omega // 2\Omega = 1\Omega$.
for $t > 0$.
$i_L(t) = B + Ae^{-\frac{t}{\tau}}$.
1). $\tau = \frac{L}{R_T} = \frac{2}{1} = 2$ [s]
2). $i_L(\infty) = B = 0$ [A].
3). $i_L(0) = i_L(0^-) = 2$ [A] $= B + A$
⇒ $A = 2$
∴ $i_L(t) = 0 + (i_L(0^-) - 0)e^{-\frac{t}{2}}$
$= 2e^{-\frac{t}{2}}$ [A].
$i_{ZS,L}(t) = 0$ [A].
$i_{ZI,L}(t) = i_L(0^-)e^{-\frac{t}{2}} = 2e^{-\frac{t}{2}}$ [A]

Consider the circuit below.

Assume the switch has been in position A for a long time and it switches to position B at time **t = 0**. Then, it switches to position A at $t = 2$.

Determine and sketch $\mathbf{i_L(t)}$. Pleas get $i_L(t)$, $t > 2$.



From #18
① $i_L(0^-) = 2A$, $t < 0$
② $i_L(t) = 2e^{-\frac{t}{2}}$ A, $0 \le t < 2$.
$i_L(2^-) = 2e^{-2/2}$ A

3). Norton
$I_N = 4/2 = 2A$
$R_T = 2\Omega$.
$i_L(t) = B' + A'e^{-t/\tau'}$, $t \ge 2$
$\tau' = 2/R_T = 1$ [s]
$i_L(\infty) = 2[A] = B'$ (L → short cbt)
$i_L(2^+) = B' + A'e^{-2/1}$
$= i_L(2^-)$.
⇒ $A' = (2e^{-1} - 2)e^2$

∴ $i_L(t)$
$= 2 + (2e^{-1} - 2)e^2 e^{-t}$
[A]